

Atharv Waingade

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PROFESSIONAL SUMMARY

AI Engineer with production experience in multi-agent LLM systems, RAG pipelines, and geometric deep learning. Delivered a Q1-ready GNN-Transformer model for 3D RNA structure prediction and deployed document intelligence systems in a commercial setting. Skilled in PyTorch, LangChain, HuggingFace, and modular agent architecture. Seeking AI/ML engineering roles focused on scalable, reasoning-capable AI systems.

TECHNICAL SKILLS

Languages: Python, NumPy, Pandas, SciPy **Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers

LLM & Agents: LangChain, RAG, Multi-Agent Systems, Prompt Engineering, Vector Retrieval **Deep Learning:** GNN, CNN, RNN, EGNN, Transformer

Specialized: Geometric Deep Learning, 3D Structure Prediction, OCR, Document AI **Evaluation:** TM-score, GDT-TS, F1-Score, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

AI / ML Engineer Intern | Code Crafters Services, Maharashtra, India

Jan 2026 – June 2026

- RAG Pipeline Architecture:** Built end-to-end retrieval-augmented generation pipelines integrating vector retrieval with LLM inference, improving chatbot contextual accuracy and reducing hallucinations in production.
- Document Intelligence:** Developed automated invoice parsing system (OCR → structured JSON + tabular output), eliminating manual data extraction workflows entirely.
- Engineered large-scale data processing pipelines for voter record datasets covering extraction, deduplication, validation, and structuring at millions-of-records scale.
- Delivered production-ready modular system designs with clean API contracts enabling rapid feature iteration and real-world deployment without breaking changes.

PROJECTS

3D RNA Structure Prediction | Q1 Publication Ready

- Designed a hybrid **GNN-Transformer architecture** with Geometric Self-Attention, Adaptive EGNN, and Rotary Positional Embeddings to predict 3D RNA atomic coordinates; achieved near-experimental TM-score and GDT-TS performance.
- Engineered dynamic mixup and synthetic structure generation augmentation pipelines; exposed model via REST API for sequence-to-structure prediction enabling downstream bioinformatics use.

Modular Multi-Agent AI Framework

- Built **role-specialized multi-agent orchestration** (Planner, Coder, Evaluator) for autonomous task decomposition with RAG-based memory for persistent context grounding across agent turns.
- Architected for composability and fault tolerance — agents fail gracefully and can be swapped without modifying the core orchestration layer.

AI Execution Intelligence & ML Debugging System

- Developed an **execution-aware ML debugging platform** that automatically traces notebook execution, tracks variable lineage, detects data leakage, distribution shift, gradient instability, and overfitting — surfacing root causes with corrective recommendations.

EDUCATION

B.Tech, Computer Engineering | Rajarambapu Institute of Technology, Walwa

2023 – 2026

Diploma, Computer Engineering | New Polytechnic Kolhapur

2020 – 2023

ACHIEVEMENTS

- RNA 3D Structure Prediction research prepared for Q1-level journal publication; achieved near-experimental structural accuracy benchmarks using geometric deep learning.
- Rare undergraduate profile combining published-quality ML research (equivariant GNNs) with production-deployed AI systems (RAG, Document AI, multi-agent pipelines).